

Supplementary Material for “Database Schemas as Quantum Fields: Relational and Document Models as a Classical Sector”

Arun Reungsilpkolkarn
School of Information Technology and Innovation, Bangkok University
arun.r@bu.ac.th

July 2026

Code and data: <https://github.com/stevezhai139/qfdb-reproducibility>

Abstract

This supplement collects everything the main paper defers: the division of labour between “field” and “quantum” (Appendix A), full proofs of the propositions with the supporting lemma and characterization remarks (B), the complete algebra of the worked temporal example (C), the S1 elicitation design with its power analysis (D), the pre-registered stored-data negative control with full results (E), the LLM-respondent pilot harness (F), and reproducibility details (G). The same document is distributed as the *companion report* in the reproducibility repository, so the main paper’s references to the companion resolve here. Every number is reproduced by the companion code.

A Why “field” and not just quantum mechanics

The paper’s three signatures (Bell/CHSH, order effects with the QQ invariant, interference) are properties of quantum *theory* generally—two qubits suffice to violate CHSH—not of field theory specifically. The word *field* is earned by a different part of the construction: data points are inserted and deleted, so the kinematics must support *variable and superposable item count*, which is exactly what second quantization provides (Fock space, $a^\dagger/a$, the number operator as cardinality, fermionic single-valued vs. bosonic multi-valued modes). A fixed-dimensional quantum system cannot even represent a collection whose cardinality is in superposition (verified in `qfdb_H1_field_modes.py`, Part 4). The division of labor is deliberate: the Fock structure supplies the *field*—it holds by construction and is not an empirical claim—while coherence and non-separability supply the *quantum*, which is empirical, and the signatures test exactly that. What remains open on the field side is stated in the paper’s scope: the continuum/Gaussian lift, and genuine field dynamics beyond the update-channel semantics. A distinctly second-quantized empirical signature (exchange-statistics interference of indistinguishable excitations, in the style of Hong–Ou–Mandel) is a natural fourth test for a future version.

B Proofs

B.1 Proposition 1 (Structure vs. state)

Write an entity state as $\rho = D + C$ in the structure basis $\{|s_k\rangle\}$, with $D = \sum_k p_k |s_k\rangle\langle s_k|$ and $C = \rho - D$.

(a) The sector $\{C = 0\}$ is closed under structure change. A definite structure is a basis vector $|s_k\rangle\langle s_k|$; a heterogeneous or merely uncertain one is a mixture $\sum_k p_k |s_k\rangle\langle s_k|$. Both are diagonal, so $C = 0$. Schema evolution acts as a stochastic matrix M on the population vector p ($p \mapsto Mp$, $M_{jk} \geq 0$, $\sum_j M_{jk} = 1$): it maps diagonal states to diagonal states and leaves C untouched. Hence no sequence of structure changes creates coherence. $\square$

(b) Which signature is carried by what. For a query outcome $|e\rangle$ the Born rule expands as

$$P(e) = \sum_k p_k |\langle e|s_k\rangle|^2 + 2 \operatorname{Re} \sum_{k < l} C_{kl} \langle e|s_k\rangle \langle s_l|e\rangle.$$

The second (interference) term is linear in C and vanishes iff $C = 0$: the S3 signature is carried by coherence alone. Every separable ρ admits a local-hidden-variable model for the CHSH settings and therefore obeys $|S| \leq 2$ (Section B.5): the S1 signature is carried by non-separability alone.

Order effects, by contrast, are a property of the *query pair*, not of C :

Lemma 1 (Commuting queries are order-invariant on every state). Let $\{P_a\}$ and $\{Q_b\}$ be projective measurements with $[P_a, Q_b] = 0$ for all a, b . Then for every state ρ the marginal of either measurement is unchanged by performing the other first.

Proof. $\Pr(b \text{ after } P) = \sum_a \operatorname{tr}(Q_b P_a \rho P_a) = \sum_a \operatorname{tr}(P_a Q_b P_a \rho) = \sum_a \operatorname{tr}(Q_b P_a \rho) = \operatorname{tr}(Q_b \rho)$, using cyclicity and $P_a Q_b P_a = Q_b P_a^2 = Q_b P_a$. The symmetric statement holds with the roles exchanged. $\square$

Conversely, an order effect requires non-commuting queries, and it can occur on a *coherence-free* state: Face A of the paper realises exactly this. Take $\rho = |0\rangle\langle 0|$ (a stored, definite record; $C = 0$ in the stored basis) and a second query Q_θ projecting at angle θ to the stored axis. Then

$$p_{AB}(0, \theta) = \cos^2 \theta \quad \text{but} \quad p_{BA}(0, \theta) = \cos^4 \theta,$$

which differ for $0 < \theta < \frac{\pi}{2}$. The first measurement creates the coherence (relative to the stored basis) that the second one feels. This is why the paper assigns order effects to the *commuting* condition of the classical sector (Definition 2), and interference and Bell violations to the state's C and non-separability respectively — one signature per sector condition. $\square$

B.2 Proposition 2 (Data-model embedding)

Build $\mathcal{H}$ by recursion: an atomic field over domain D is a single mode $\mathbb{C}^{D \cup \{\perp\}}$; an ordered array of element type τ is the sequence space $\bigoplus_k \mathcal{H}_\tau^{\otimes k}$, and a set/multiset is its symmetric subspace; a nested document is the tensor product of its sub-fields.

Proof. Each record vector $|t\rangle = \bigotimes_f |t_f\rangle$ is by construction a *product vector* across the field factors ($\perp$ -padding for heterogeneous collections keeps this true for document data). A relation or collection r is the state $\rho_r = \frac{1}{|r|} \sum_{t \in r} |t\rangle\langle t|$, a convex combination of product states — **separable by the definition of separability**; no positive-partial-transpose or negativity criterion is invoked, so the bound-entanglement counterexamples to “PPT $\Rightarrow$ separable” (which exist from dimension 2×4 and 3×3 up) are irrelevant to the argument. ρ_r is also diagonal in the record basis, so $C = 0$, and selection/membership projectors $\{|t\rangle\langle t|\}$ commute. By Proposition 1 the state lies in the classical sector. Nothing in the argument depends on whether a field factor is a single mode (Codd’s 1NF) or an enriched sequence/nesting space, so the relational and document models occupy one and the same diagonal, separable sector. *Remark:* in the paper’s 2×2 example $\{(a, x), (b, y)\}$, quoting “negativity 0” is conclusive, because PPT $\Leftrightarrow$ separable holds in dimensions 2×2 and 2×3 (Horodecki); the caveat above applies only to higher dimensions. $\square$

B.3 Proposition 3 (Temporal order-invariance)

Proposition 1 (restated). A set of temporal queries over an entity’s history is order-invariant — its outcomes admit a single classical joint distribution, every marginal independent of query order — iff the histories’ decoherence functional obeys $\text{Re } D(\alpha, \beta) = 0$ for $\alpha \neq \beta$. Logging every intermediate stage the queries distinguish is sufficient; it is necessary exactly when the unlogged alternatives carry coherence.

Proof. The equivalence between the existence of a single classical joint distribution (hence order-independent marginals) and the weak consistency condition $\text{Re } D(\alpha, \beta) = 0$ is the standard consistency theorem of the consistent-histories formalism (Griffiths). For the second clause: a logged intermediate stage inserts a projective measurement at that stage, which dephases the history alternatives and forces $\text{Re } D(\alpha, \beta) = 0$ — sufficiency. Necessity fails in general: a state already diagonal on the alternatives (a classical-sector state, Definition 2 of the paper) satisfies the condition with no logging at all — which is precisely why an ordinary unlogged classical database is order-safe. For states carrying coherence between the alternatives, a generic relative phase gives $\text{Re } D(\alpha, \beta) \neq 0$, and logging is then exactly what restores order-safety. Hence: *once histories carry coherence*, logging granularity is the order-safety boundary. $\square$

B.4 Characterization of the sector: one-way guarantees

The paper deliberately states the characterization as one-way guarantees. Two remarks make precise why the naive “iff” would be false. (i) $|S| \leq 2$ does not imply separability: Werner states in 2×2 are entangled yet satisfy the CHSH bound for all measurement settings (and in a visibility range even admit a full local-hidden-variable model). $S > 2$ *certifies* non-separability; $S \leq 2$ certifies nothing. (ii) For a fixed state, order-invariance of the realised queries does not imply the query algebra commutes. The correct statement quantifies over states (Lemma 1): a commuting family is order-invariant on *every* state, and any observed signature beyond its classical bound certifies that the (state, queries, histories) triple has left the sector.

B.5 CHSH: the two-line classical bound and both ends

If each item carries predetermined ± 1 answers a, a', b, b' under both settings of each side, then $a(b + b') + a'(b - b')$ has $b \pm b' \in \{0, \pm 2\}$, one of the two brackets vanishing, so the expression

is ± 2 and $|S| = |\mathbb{E}[\cdot]| \leq 2$. Mixtures (noisy or contextual shared-latent strategies) are convex combinations and inherit the bound. The companion script `qfdb_commoncause_bound.py` enumerates all 16 deterministic common-cause strategies (all lie in $[-2, 2]$), samples 2×10^5 random mixtures ($\max |S| = 1.4514$), and evaluates the Bell state with CHSH-optimal settings ($S = 2\sqrt{2} \approx 2.8284$).

C Worked example: full algebra

Lifecycle: *calf* $\rightarrow$ *steer* $\rightarrow$ *ox* (one male animal; the paper’s running example).

Face A (order effect on a stored record). State $|\neg\text{vacc}\rangle$ stored at plane t ; mutant-susceptibility query = projector at angle θ . Asked alone: $\Pr(\neg\text{vacc}) = 1.000$. Asked after the mutant query: $\Pr = \cos^4 \frac{\theta}{2} + \sin^4 \frac{\theta}{2}$ in Bloch convention, giving 1.000, 0.875, 0.750, 0.625, 0.500 at $\theta = 0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$ — an order effect up to 0.500, while the Wang–Busemeyer QQ invariant is 0.000 at every θ (`qfdb_order_effect_mutant.py`).

A classical disturbance model tuned to the same order effect — B-first fully redraws A (50/50), $B \sim \text{Bernoulli}(r)$ — gives

$$\text{QQ} = [p(A_y B_y) + p(A_n B_n)] - [p(B_y A_y) + p(B_n A_n)] = r - \frac{1}{2},$$

i.e. $\text{QQ} = 0.10, 0.20, 0.30$ at prevalence $r = 0.6, 0.7, 0.8$: zero only at one fine-tuned base rate, whereas the quantum value is parameter-free. Pinning that residual is what S1 targets.

Face B (unlogged history). Histories {through-steer, not-through-steer} with amplitudes a, b and relative phase φ . Unlogged: $\Pr(\text{ox now}) = |a + be^{i\varphi}|^2$ ranges over $[(|a| - |b|)^2, (|a| + |b|)^2]$; the observed swing $[0.000, 0.444]$ forces $|a| = |b|$ with $4|a|^2 = 0.444$, i.e. $|a|^2 = 1/9$. Logged: the decoherence functional is dephased, $\Pr = |a|^2 + |b|^2 = 2/9 = 0.222$, phase-blind. The decoherence functional’s off-diagonal magnitude is 0.111 unlogged and 0.000 logged (`qfdb_consistent_histories.py`).

Complementarity (supporting check). With visibility V and which-structure distinguishability D , $V^2 + D^2 \leq 1$ (Englert). Reading records as a which-structure detector: pinning records ($D \rightarrow 1$) sharpens the schema; leaving them indefinite keeps it field-like — a quantum form of schema-on-write vs. schema-on-read. The inequality holds for any two-outcome system, classical bits included, so it is a coherence check of the framework, not a non-classicality test.

D S1 elicitation design (full)

Parties = two category systems describing the same products. Settings: $A = \{\text{Electronics, Furniture}\}$, $A' = \{\text{Gift, Tool}\}$; $B = \{\text{Gadget, Decor}\}$, $B' = \{\text{Toy, Hardware}\}$. One respondent sees *one* item and *one* setting pair (between-subjects, randomised — the free-choice condition), makes a forced-choice category judgment on each side, coded ± 1 . Acceptance criterion: violation iff the 95% bootstrap CI on S lies above 2 *and* the no-signaling (marginal-law) check passes; marginal shifts with the distant setting read as contextuality-with-signaling, not a Bell violation. A violation certifies non-classical structure in correspondence *judgments* (not in the stored schema-field — the Level-2 distinction). Power: for an effect near the Aerts–Sozzo magnitude

($S = 2.42$, $N = 81$), on the order of 10^2 respondents per setting pair; a pre-registered interim look allows early stopping. The Aerts–Sozzo line has been criticised for marginal-selectivity violations (Dzhafarov–Kujala); the no-signaling gate above is the design answer. Instrument, tally and no-signaling scripts: `s1_elicitation/`.

E Pre-registered stored-data negative control

Pre-registration frozen before evaluation (`s1_stored/PREREGISTRATION.md`, sha256 prefix 72cf065266be0124). Prediction: stored two-store correlations are common-cause, hence $S \leq 2$ everywhere, adversarial settings included; for per-side-deterministic responses this is a theorem (the shared item is the latent variable of Section B.5), so the run validates the instrument end-to-end and measures the margin.

Data. Magellan/ditto entity-matching benchmarks; items = ground-truth matched pairs; rule library (numeric-median, length, top-8 token rules) built on TRAIN per side; evaluation on held-out VALIDUTEST. Locality by construction: each side’s ± 1 response is a function of that side’s record and setting only. **Analyses.** (A) natural settings, same items; (B) adversarial: exhaustive search over setting pairs maximising $|S|$ on TRAIN, evaluated once held-out; (C) between-subjects dry run (four disjoint item groups) with no-signaling deltas. **Self-tests** (pass before data): Bell 2.8284; engineered classical correlations attain $S = 2.000$ exactly (the instrument is sharp at the bound); product state ≈ 0 .

Table 1: Held-out results (bootstrap 95% CIs, 10^4 resamples). No CI lies above 2: the classical-sector prediction is verified on every analysable pair, while the same pipeline’s entangled control reaches 2.8284.

dataset	n	S_{nat}	$S_{\text{adv}}^{\text{train}}$	$S_{\text{adv}}^{\text{held-out}}$	95% CI	NS Δ
Amazon–Google	468	+1.026	+1.966	+1.974	[+1.940, +2.000]	0.154
DBLP–ACM	888	−0.189	+2.000	+1.995	[+1.986, +2.000]	0.189
DBLP–GoogleScholar	2140	−0.456	+1.988	+1.983	[+1.972, +1.993]	0.037
Fodors–Zagats	44	−0.364	+2.000	+2.000	[+2.000, +2.000]	0.364
Walmart–Amazon	386	−0.166	+1.979	+1.938	[+1.886, +1.979]	0.103
iTunes–Amazon	54	+0.815	+2.000	+2.000	[+2.000, +2.000]	0.319

Beer (≈ 20 held-out matches) is excluded by a ≥ 30 -item practicality rule adopted at analysis time and disclosed here (it was not part of the frozen pre-registration); all other pairs are reported without selection. The between-subjects no-signaling deltas scale with per-group n — the direct lesson for the human study’s sample size. Adversarial search saturates the deterministic bound on strongly-correlated pairs (exactly 2.000) and crowds it elsewhere (1.94–1.99): real integration data pushes against the classical ceiling but cannot pass it, exactly as the embedding requires. Reproduce with `s1_stored/s1_stored_negative_control.py` (fetch commands in the pre-registration).

F LLM-respondent pilot harness

`s1_elicitation/llm_harness/` implements the S1 protocol with a language model as synthetic respondent: one API call = one respondent (fresh context; between-subjects), one setting pair per call, neutral catalogue-filing wording (no physics vocabulary), option order counterbalanced,

temperature > 0 with a reported grid, open-weights model preferred and version pinned, full JSONL logs, and the same acceptance criterion ($CI > 2$ *and* no-signaling). A violation would certify non-classical structure in the *model's* judgment statistics — not in human cognition and not in the schema-field; the pilot's roles are instrument validation and an effect-size prior for the human study. A `--dry-run` classical simulated judge exercises the analysis path without any API.

G Reproducibility

Reference environments. All results in this report were reproduced digit-for-digit (including bootstrap CIs, via a fixed PCG64 seed) on two independent machines: (a) a MacBook Pro (Apple M4, 24 GB RAM, 512 GB internal SSD + 1 TB Thunderbolt external SSD) running macOS Tahoe 26.x with NumPy 2.2.6, and (b) a Debian Linux aarch64 container with Python 3.10 and NumPy 2.2.6. No GPU or quantum hardware is used; the full negative-control run completes in under a minute on either machine.

All scripts are NumPy-only (the LLM harness is stdlib-only). `qfdb_signatures_verify.py`, `qfdb_order_effect_mutant.py`, `qfdb_consistent_histories.py`, `qfdb_defs_verify.py`, `qfdb_commoncause.bo`, `qfdb_H1_field_modes.py` reproduce every computed number in the main paper; `s2_calibration/` reproduces the 72-survey calibration (order-effect mean 0.087, range 0.03–0.58; QQ mean $+0.001 \pm 0.036$, $t(71) = 0.25$, $p = 0.80$); `s1_stored/` reproduces the table above.